



CAPSTONE PROJECT

Predicting Smoking Status from Biometric Health Data

A machine learning approach to actuarial longevity risk segmentation

INTRODUCTION

About This Project

CONTEXT

A biometric health dataset of 55,692 individuals used to predict smoking status from clinical blood test results and body measurements.

55,692	27	0	63 / 37
RECORDS	FEATURES	MISSING VALUES	CLASS SPLIT %

Target: smoking (1 = Smoker, 0 = Non-Smoker)
Key features: hemoglobin, liver enzymes, lipid panel, blood pressure, BMI, gender.

PROBLEM FRAMING

Problem Statement + Aim

THE PROBLEM

Insurers and pension funds can't reliably identify smokers from biometric data alone, leading to mispriced premiums and underestimated mortality risk.

AIM

Build a binary classification model to predict smoking status from biometric and clinical measurements, enabling individual-level actuarial risk segmentation.

SUCCESS METRICS

Recall ≥ 0.75 for the smoker class **≥ 0.75**

F1-Score ≥ 0.75 (balanced precision & recall) **≥ 0.75**

DATA SUMMARY

What Data Are We Using?

Feature	Description	Corr. w/ Smoking
hemoglobin	Blood hemoglobin level (g/dL)	+0.40 — strongest
gender	Biological sex (M/F)	M: 55.4% vs F: 4.2% smoking rate
triglyceride	Blood lipid level (mg/dL)	+0.25
liver_risk_score	Engineered: avg of Gtp, ALT, AST	+0.24 composite
HDL	Good cholesterol (mg/dL)	-0.18
BMI	Engineered: weight / height ²	+0.15

Three features engineered: BMI (replaces height + weight), Pulse Pressure (systolic – diastolic), Liver Risk Score (normalized avg of Gtp, ALT, AST).

EXPLORATORY DATA ANALYSIS

Key Findings

GENDER GAP

55.4%

male smoking rate

4.2%

female smoking rate

~13x difference — confirmed by Chi-Square ($p \approx 0.000$)

ADDITIONAL INSIGHTS

- Smokers are younger on average (peak: 30–45)
- Tartar presence statistically associated with smoking ($p < 0.05$)
- Triglycerides higher (+0.25), HDL lower (-0.18) in smokers

Hemoglobin

+0.40

Strongest biomarker — body produces more due to reduced oxygen from smoking

Liver Enzymes (Gtp)

+0.24

9% outlier rate — toxic liver load concentrated in smokers

HDL Cholesterol

-0.18

Smoking suppresses protective cholesterol

MODELING APPROACH

How We Built the Models

FEATURE ENGINEERING

BMI, Pulse Pressure, Liver Risk Score

DROPPED COLUMNS

height, weight, waist, systolic, relaxation, Gtp, ALT, AST

ENCODING

gender (M=1, F=0), tartar (Y=1, N=0)

SCALING

RobustScaler for high-outlier columns, StandardScaler for the rest

TRAIN/TEST SPLIT

80/20, stratified by target, random_state=42

CLASS IMBALANCE

class_weight='balanced' applied to LR and Random Forest

MODELS

Logistic Regression · KNN · Random Forest

All improved using GridSearchCV optimizing for Recall

MODEL 1

Logistic Regression

GridSearchCV over $C \in \{0.01, 0.1, 1, 10, 100\}$ with `class_weight='balanced'` — Best $C = 1$

EQUATION

$P(\text{Smoker}) = \sigma(\beta_0 + \beta_1 \cdot \text{gender} + \beta_2 \cdot \text{hemoglobin} + \beta_3 \cdot \text{triglyceride} + \beta_4 \cdot \text{liver_risk_score} + \dots)$

PERFORMANCE

0.92

Recall (Smoker)

0.70

F1-Score (Smoker)

0.71

Accuracy

Recall target met (0.92). F1 is below 0.75 due to the precision-recall tradeoff from `class_weight='balanced'`.

MODEL 2

K-Nearest Neighbors (KNN)

GridSearchCV over $k \in \{3, 5, 7, 9, 11, 15\}$ with Euclidean and Manhattan metrics — Best $k = 15$, Euclidean.

PERFORMANCE

0.61

Recall (Smoker)

0.62

F1-Score (Smoker)

0.72

Accuracy

Worst performing model. KNN struggles with 24 features — high dimensionality makes distance calculations less meaningful (curse of dimensionality). Neither target met.

MODEL 3

Random Forest

GridSearchCV with class_weight='balanced' — Best: max_depth=10, min_samples_split=5, n_estimators=200

PERFORMANCE

0.93 ✓

Recall (Smoker)

0.71

F1-Score (Smoker)

0.72

Accuracy

TOP FEATURES

1. gender — 2. hemoglobin — 3. triglyceride — 4. liver_risk_score — 5. LDL

Highest Recall (0.93). Handles non-linear relationships and is robust to the outliers present in clinical features.

What This Model Delivers

Even without meeting both targets on F1, this model delivers real value to insurers and pension funds by identifying smokers more reliably than demographic data alone.

Risk Flagging

A Recall of 0.93 means 93% of smokers are correctly identified — reducing the risk of mispriced life insurance premiums for high-risk individuals.

Underwriting Support

Model predictions can be used alongside medical records to flag individuals for further underwriting review, rather than replacing human judgment.

Portfolio Segmentation

Enables insurers to segment policyholders by predicted smoking risk, allowing more accurate mortality assumptions in pension fund pricing.

Fraud Detection

Discrepancies between predicted smoking status and declared status can flag potential misrepresentation in insurance applications.

The model is not a replacement for underwriting expertise — it is a decision-support tool that makes risk segmentation faster and more data-driven.

CONCLUSION

Model Comparison

Model	Recall (Smoker)	F1-Score (Smoker)	Accuracy	Recall Target Met?
Logistic Regression	0.92	0.70	0.71	Yes ✓
KNN	0.61	0.62	0.72	No ✗
Random Forest	0.93	0.71	0.72	Yes ✓

BEST MODEL: RANDOM FOREST

Random Forest achieved the highest Recall (0.93) and balanced performance across metrics. It outperforms the others because it handles non-linear relationships, is robust to outliers, and captures feature interactions that simpler models can't.

Next steps: Try XGBoost/LightGBM, use SHAP values for interpretability, collect smoking duration data.